YDTR: Infrared and Visible Image Fusion via Y-Shape Dynamic Transformer

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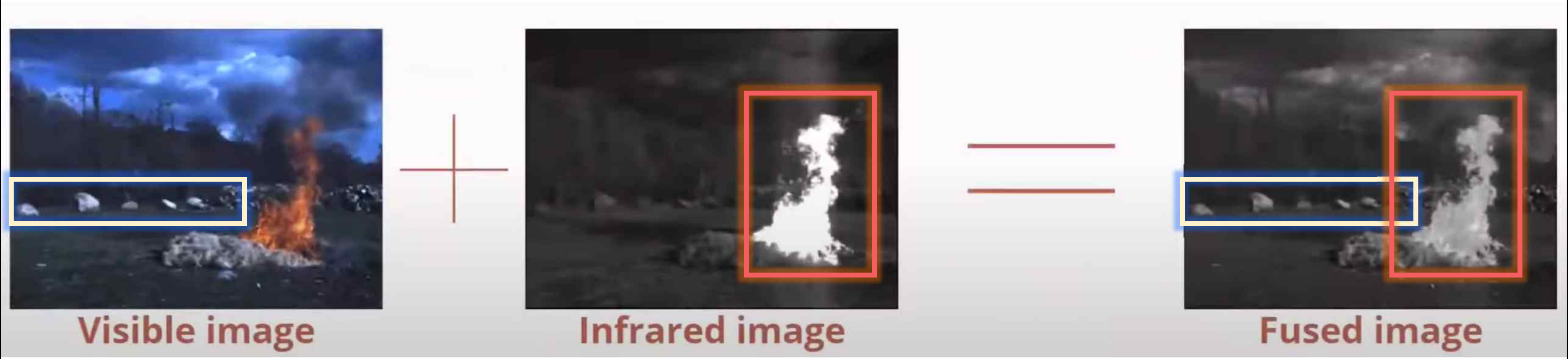
Introduction

IR-Vision Image fusion



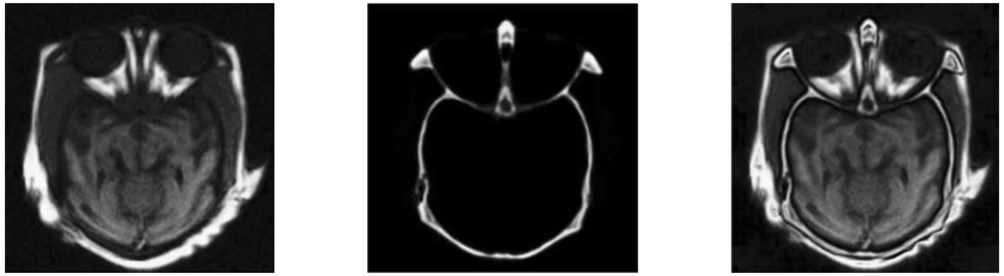
Jorge Francisco Ciprián Sánchez, "Deep Learning for visible-infrared image fusion and semantic segmentation of wildfire imagery" (https://youtu.be/K0nkNt_yuLg?si=IMpwykeQ3Ck9VYAI)

Introduction

IR-Vision Image fusion

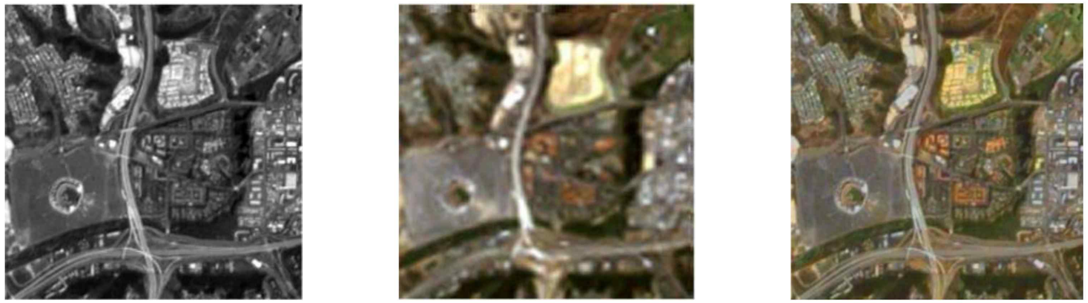
(a) (b) (c)

Robot vision field



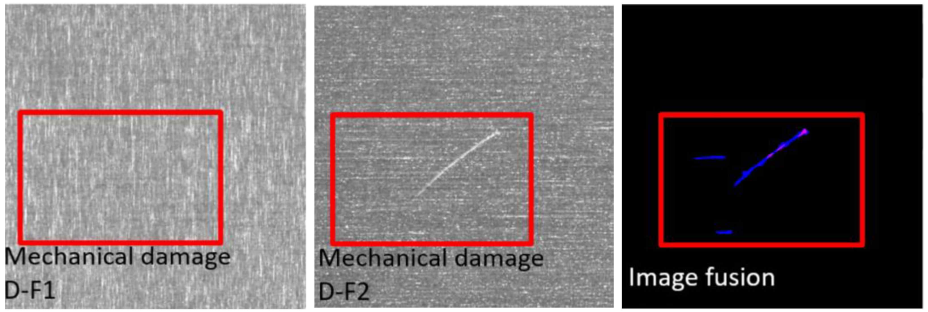
(a) (b) (c)

Medical Imaging



(a) (b) (c)

Agricultural remote sensing



(a) (b) (c)

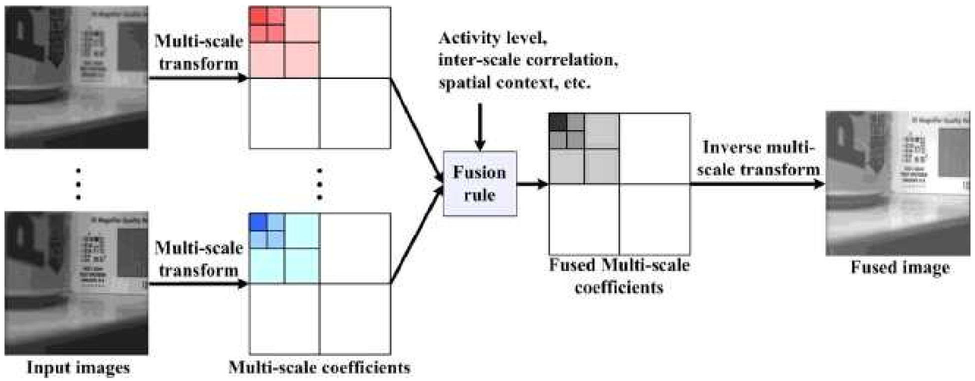
Industrial defect detection

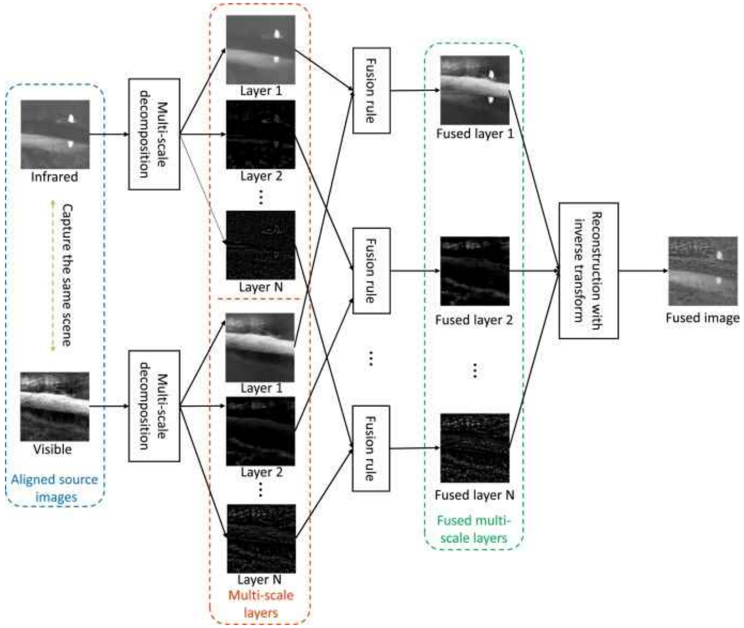
Ma, W.; Wang, K.; Li, J.; Yang, S.X.; Li, J.; Song, L.; Li, Q. Infrared and Visible Image Fusion Technology and Application: A Review. Sensors 2023, 23, 599. https://doi.org/10.3390/s23020599

Introduction

Fusion Methods – Multi-scale transform

- Decomposition method:
  Pyramid, wavelet, curvelet, contourlet, shearlet, edge-preserving decomposition, and etc.
- Fusion rules:
  Coeffieicnt, window, choose-max, optimization based method, weighted-average, and etc.



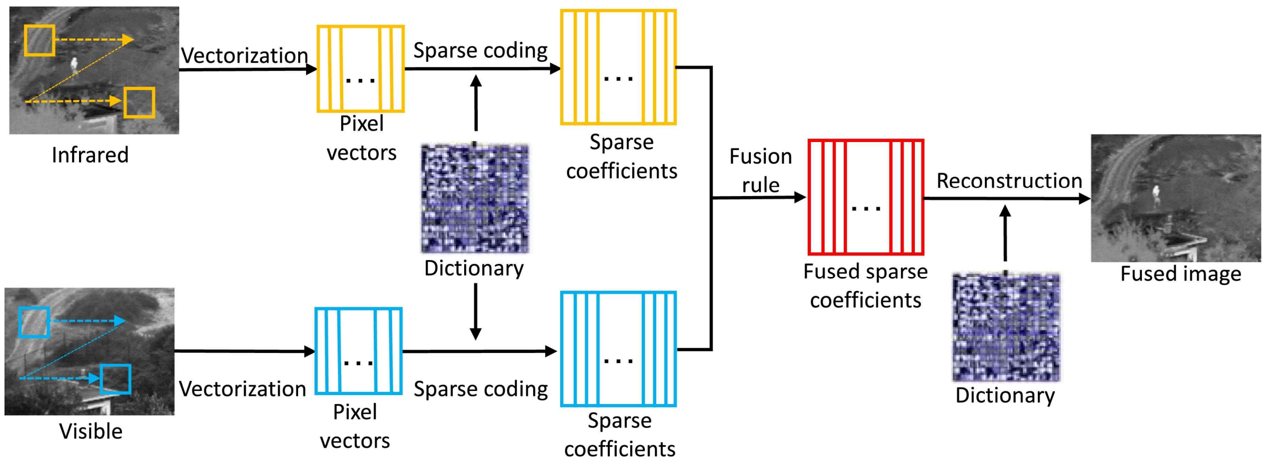


S., Kang, X., Fang, L., Hu, J., & Yin, H. (2017). Pixel-level image fusion: A survey of the state of the art. Information Fusion, 33, 100–112. http://dx.doi.org/10.1016/j.inffus.2016.05.004.
J. Ma, Y. Ma, C. Li, Infrared and visible image fusion methods and applications: A survey, Inf. Fusion 45 (2019) 153–178.

Introduction

Fusion Methods – Sparse representation

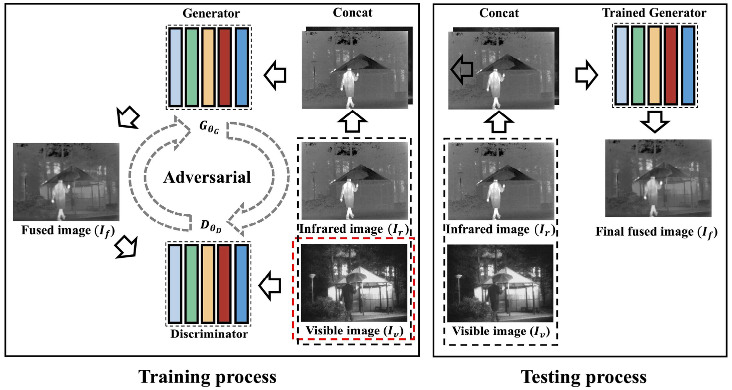
- Over-complete dictionary:
  fixed-basis, learning based method(K-means, SVD)
- Sparse coding:
  orthogonal matching pursuit, joint sparse representation model, convolutional sparse representation
- Fusion rule:
  choose-max, weighted average, etc.



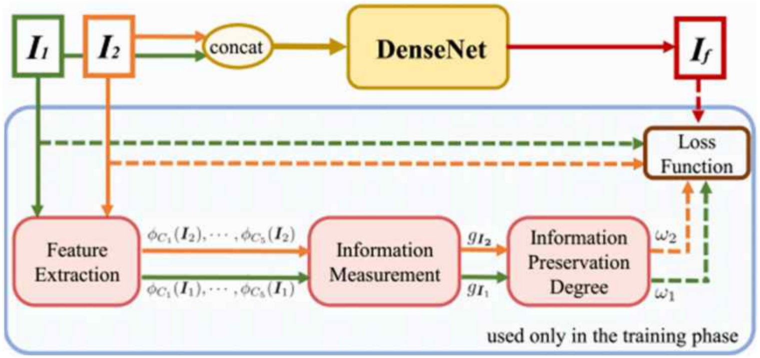
S., Kang, X., Fang, L., Hu, J., & Yin, H. (2017). Pixel-level image fusion: A survey of the state of the art. Information Fusion, 33, 100–112. http://dx.doi.org/10.1016/j.inffus.2016.05.004.
J. Ma, Y. Ma, C. Li, Infrared and visible image fusion methods and applications: A survey, Inf. Fusion 45 (2019) 153–178.

Introduction

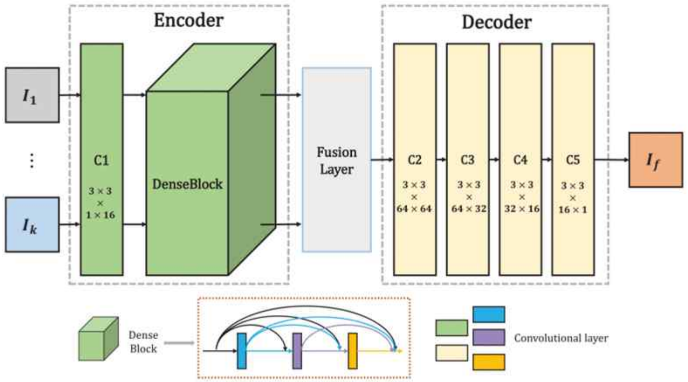
Fusion Methods – Deep-learning based



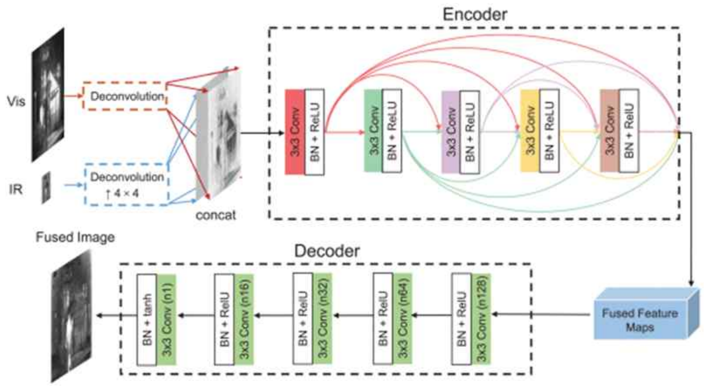
FusionGAN



U2 Fusion



DenseFuse

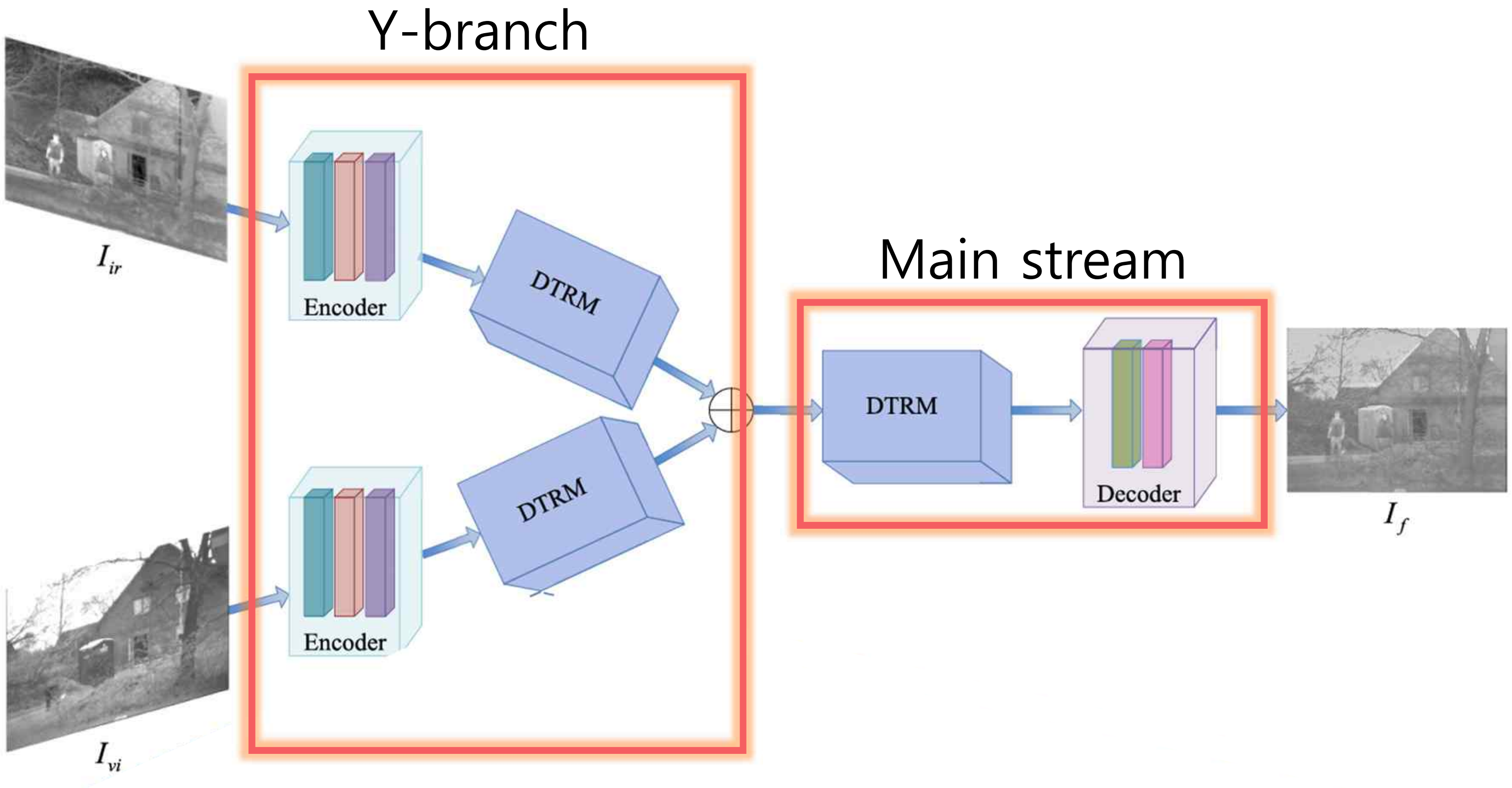


DDcGAN

Question

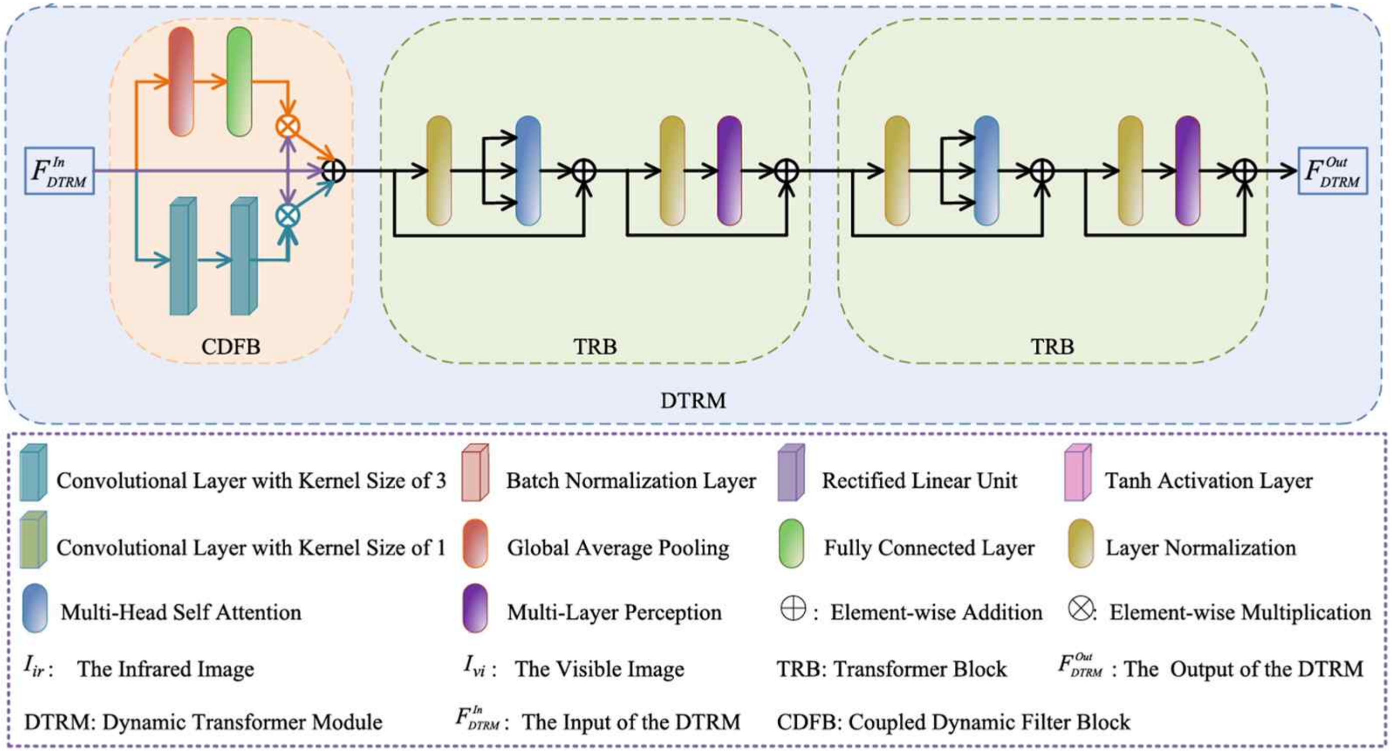
YDTR

YDTR Structure



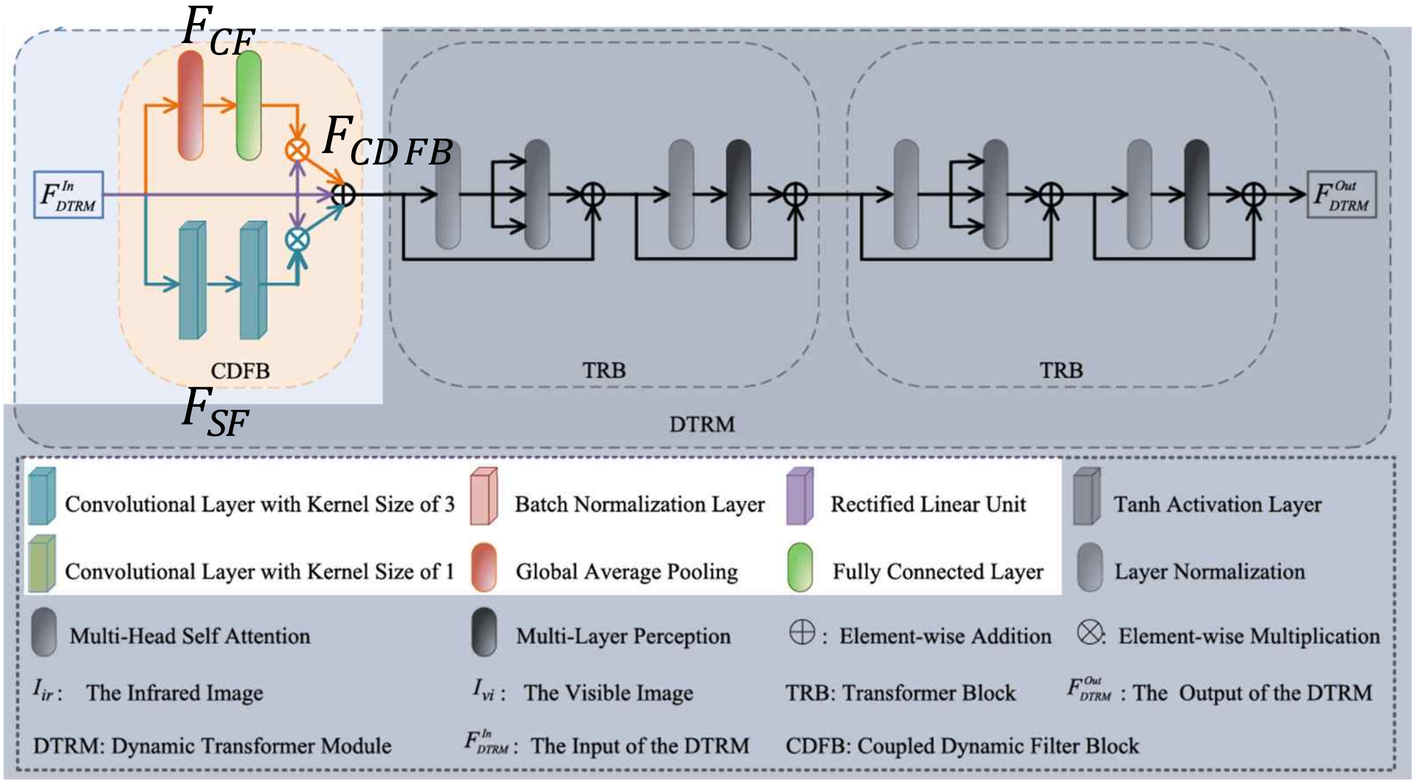
YDTR

Dynamic Transformer Module



YDTR

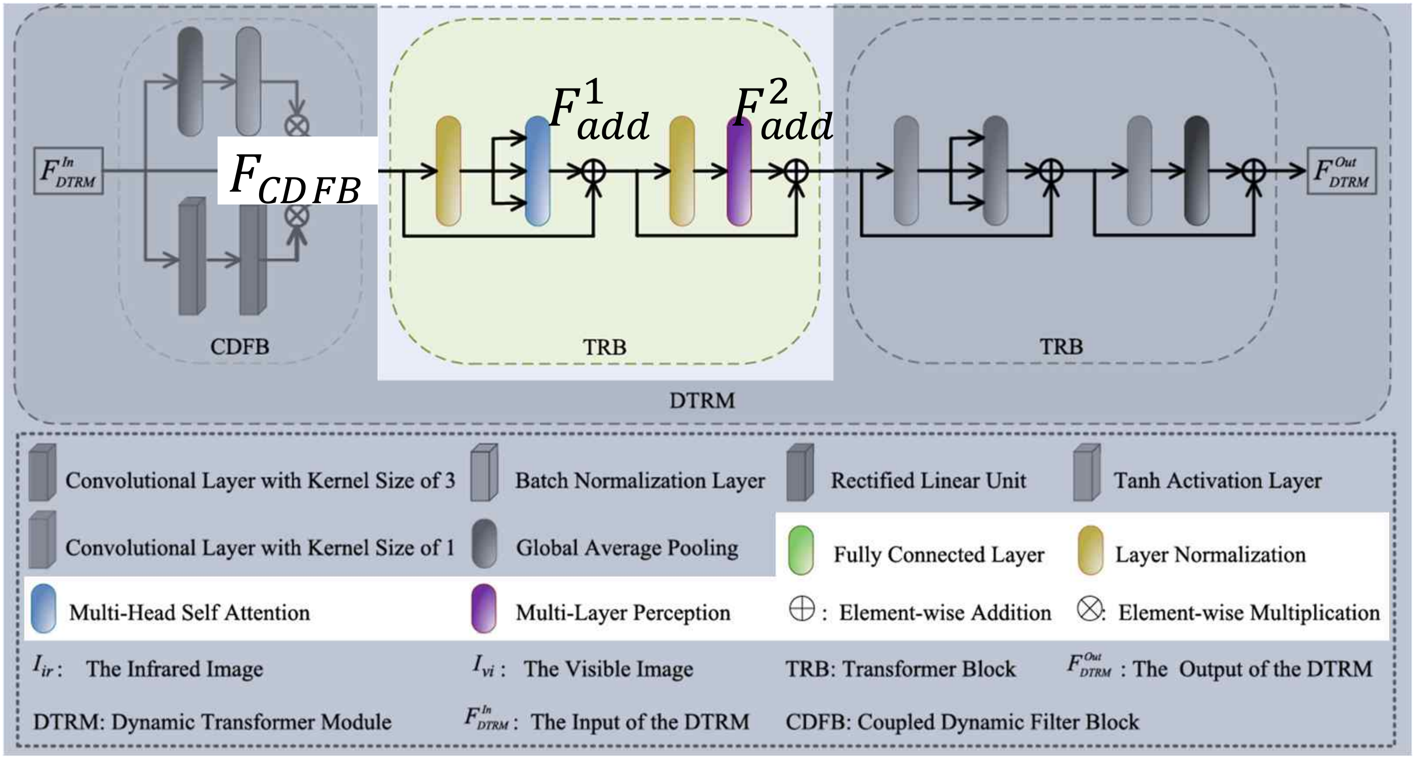
Dynamic Transformer Module



$$F_{CDFB} = F_{CF} \cdot F^{In}_{DTRM} + F^{In}_{DTRM} + F^{In}_{DTRM} \cdot F_{SF}, \quad (1)$$

YDTR

Dynamic Transformer Module



$$F_{\text{add}}^{1} = F_{CDFB} + MSA\left(LN\left(F_{CDFB}\right)\right) \qquad F_{add}^{2} = F_{add}^{1} + MLP\left(LN\left(F_{add}^{1}\right)\right),$$

YDTR

Loss Function

$$L = \underbrace{L_{SSIM}(I_f, I_s)}_{\text{Structural term}} + \underbrace{L_{SF}(I_f, I_s)}_{\boxed{\text{Spatial frequency term}}}$$

YDTR

Loss Function

$$L = \underbrace{L_{SSIM}\left(I_f, I_s\right)}_{\text{Structural term}} + \underbrace{L_{SF}\left(I_f, I_s\right)}_{\boxed{\text{Spatial frequency term}}}$$

$$L_{SSIM}\left(I_f, I_s\right) = 1 - SSIM\left(I_f, I_s\right)$$

$$SSIM\left(I_f, I_s\right) = \frac{\left(2\mu_s\mu_f + C_1\right)\left(2\sigma_{sf} + C_2\right)}{\left(\mu_s^2 + \mu_f^2 + C_1\right)\left(\sigma_s^2 + \sigma_f^2 + C_2\right)}$$

YDTR

Loss Function

$$L = \underbrace{L_{SSIM}\left(I_f, I_s\right)}_{\text{Structural term}} + \underbrace{L_{SF}\left(I_f, I_s\right)}_{\boxed{\text{Spatial frequency term}}}$$

$$L_{SF}\left(I_f, I_s\right) = \left\| SF\left(I_f\right) - SF\left(I_s\right) \right\|_2$$

$$SF = 1 - \sqrt{Hor^2 + Ver^2}$$

YDTR

Loss Function

$$L = \underbrace{L_{SSIM}\left(I_f, I_s\right)}_{\text{Structural term}} + \underbrace{L_{SF}\left(I_f, I_s\right)}_{\boxed{\text{Spatial frequency term}}}$$

$$\begin{aligned} L = {} & L_{SSIM}\left(I_f, I_{ir}\right) + \alpha \cdot L_{SSIM}\left(I_f, I_{vi}\right) \\ & + \beta \cdot L_{SF}\left(I_f, I_{ir}\right) + \gamma \cdot L_{SF}\left(I_f, I_{vi}\right), \end{aligned} \tag{13}$$

where α, β, and γ are three trade-off parameters that control the balance of the loss function. The impacts of these three factors are analyzed in Section IV-C.

Question

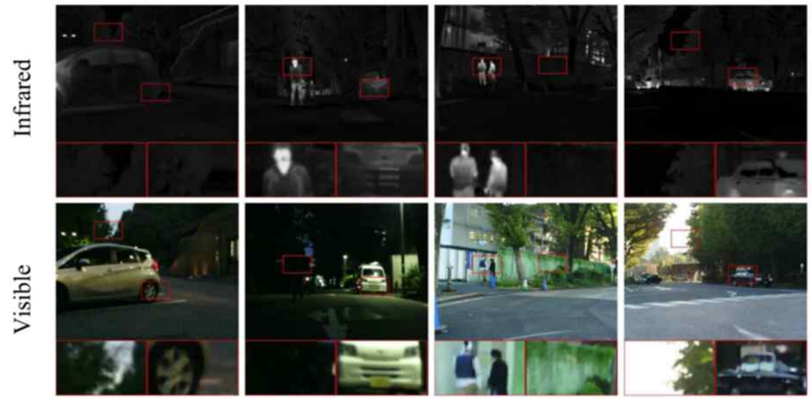
Experiment and results
Dataset

Infrared - Visible

TNO dataset

RoadScene dataset

Infrared – RGB(Visible)



Multi-Spectral Road Scenarios(MSRS) dataset

Multi-Focus Image

Left-focus Right-focus

Experiment and results
Evaluation metrics

- **Normalized mutual information Q_{MI}**

$$Q_{MI} = \frac{I(X,Y)}{\max\{H(X),H(Y)\}}, \quad H = entropies \ \ of \ \ X,Y, \quad I = cross \ \ entropy \ \ between \ \ the \ \ X,Y$$

- **Nonlinear correlation information entropy Q_{NCIE}**

$$Q_{NCIE} = 1 - H_{RN}, \quad H_{RN} = nonlinear \ \ joint \ \ entropy$$

$$\begin{aligned} P_p &= max\left(C^p_{1f}, C^p_{2f}, C^p_{mf}\right) \\ P_M &= max\left(C^M_{1f}, C^M_{2f}, C^M_{mf}\right) \\ P_m &= max\left(C^m_{1f}, C^m_{2f}, C^m_{mf}\right) \end{aligned}$$

- **Phase congruency based feature-wise metric Q_P**

$$Q_P = (P_p)^{\alpha}(P_M)^{\beta}\ (P_m)^{\gamma}$$

$$C^k_{xy} = \frac{\sigma^k_{xy} + C_k}{\sigma^k_x \sigma^k_y + C_k}$$

- **Multi-scale structural similarity index $MS-SSIM$**

$$\mathrm{SSIM}(\mathbf{x},\mathbf{y}) = \frac{(2\mu_x\mu_y + C_1)(2\sigma_{xy} + C_2)}{(\mu_x^2 + \mu_y^2 + C_1)(\sigma_x^2 + \sigma_y^2 + C_2)}, \quad \mathrm{SSIM}(\mathbf{x},\mathbf{y}) = [l_M(\mathbf{x},\mathbf{y})]^{\alpha_M} \cdot \prod_{j=1}^{M} [c_j(\mathbf{x},\mathbf{y})]^{\beta_j} [s_j(\mathbf{x},\mathbf{y})]^{\gamma_j}$$

- **Chen-Varshney metric Q_{CV}**

$$Q = \frac{\sum_{l=1}^{N} \sum_{l=1}^{L} (\lambda(X_I^{W_l}) D(X_I^{W_l}, X_F^{W_l}))}{\sum_{l=1}^{N} \sum_{l=1}^{L} (\lambda(X_I^{W_l}))}$$

Experiment and results

Ablation Study – YDTR Network

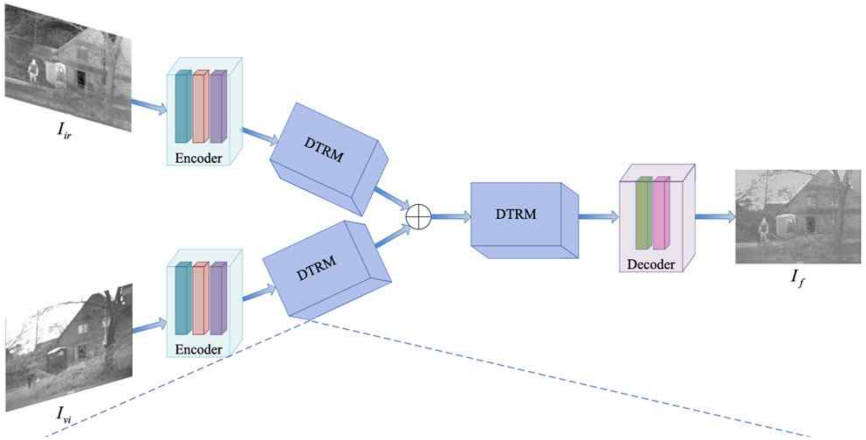


TABLE VI
OBJECTIVE EVALUATION RESULTS OF THE PROPOSED METHOD WITH DIFFERENT NETWORK STRUCTURES

	Q_{MI} [46] ↑	Q_{NCIE} [47] ↑	Q_P [48] ↑	$MS\text{-}SSIM$ [49] ↑	Q_{CV} [50] ↓
Single Path	0.3733	0.8045	0.3017	0.6279	124.3958
Concat	0.3983	0.8067	0.329	0.6822	99.6134
w/o CDFB	0.3306	0.8054	0.2379	0.6747	102.5727
w/o TRB	0.3605	0.806	0.2507	0.6782	96.468
w/o DTRM	0.288	0.8046	0.1426	0.4665	126.1872
w/o DTRM_1	0.3586	0.8064	0.2352	0.4704	109.2994
w/o DTRM_2	0.3552	0.8063	0.2307	0.5377	105.1364
w/o DTRM_3	0.3707	0.8063	0.2681	0.6812	106.703
w/o $\text{DTRM}_{2\&3}$	0.2945	0.8057	0.2246	0.4697	118.25
w/o $\text{DTRM}_{1\&3}$	0.2943	0.806	0.2069	0.4693	119.6726
w/o $\text{DTRM}_{1\&2}$	0.3503	0.8059	0.2231	0.4696	120.6865
YDTR	**0.4401**	**0.8078**	**0.3938**	**0.6884**	**69.2637**

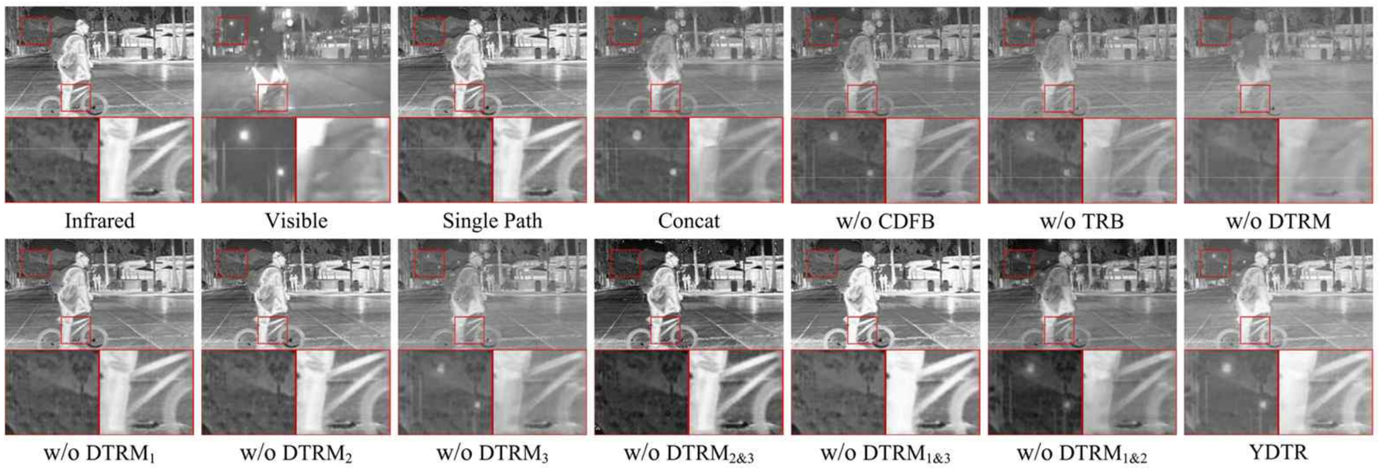


Fig. 7. A pair of source images and their corresponding fusion results of the proposed method with different network structures.

Ablation Study – Loss function

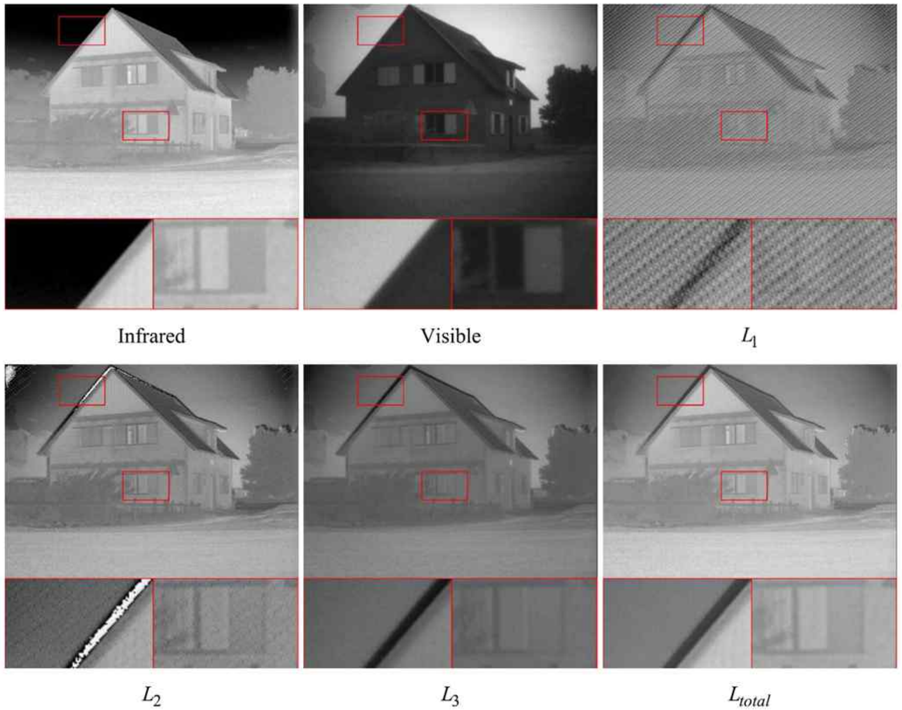


Fig. 3. A pair of source images and their corresponding fusion results of the proposed method with different loss functions.

TABLE II
OBJECTIVE EVALUATION RESULTS OF THE PROPOSED METHOD WITH DIFFERENT LOSS FUNCTIONS

Metrics	L_1	L_2	L_3	L_{total}
Q_{MI} [46] ↑	0.1237	0.3240	0.4013	**0.4401**
Q_{NCIE} [47] ↑	0.8032	0.8055	0.8065	**0.8078**
Q_P [48] ↑	0.0514	0.2367	0.3535	**0.3938**
$MS\text{-}SSIM$ [49] ↑	0.4726	0.6698	0.6702	**0.6884**
Q_{CV} [50] ↓	96.7566	92.992	75.6981	**69.2637**

$$L_1 = L_{SF}(I_f, I_s). \tag{14}$$

Similarly, we remove L_{SF} from L_{total} to demonstrate its necessity, and rewrite L_{total} as

$$L_2 = L_{SSIM}(I_f, I_s). \tag{15}$$

To further illustrate the irreplaceability of the proposed L_{SF}, we replace L_{SF} with a gradient operator, and redefine L_{total} as

$$L_3 = L_{SSIM}(I_f, I_s) + L_G(I_f, I_s), \tag{16}$$

where $L_G(I_f, I_s)$ is the gradient term, which is expressed as

$$L_G(I_f, I_s) = \|\nabla I_f - \nabla I_s\|_2, \tag{17}$$

Experiment and results

Ablation Study – Weight parameter

$$L = L_{SSIM}(I_f, I_{ir}) + \alpha \cdot L_{SSIM}(I_f, I_{vi}) + \beta \cdot L_{SF}(I_f, I_{ir}) + \gamma \cdot L_{SF}(I_f, I_{vi})$$

α

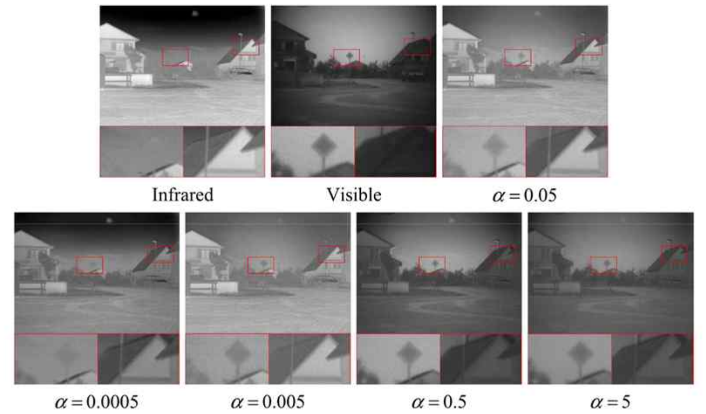


Fig. 4. A pair of source images and their corresponding fusion results of the proposed method using different values of α in the loss function (β and γ are fixed as 0.0006 and 0.00025, respectively).

TABLE III
OBJECTIVE EVALUATION OF THE PROPOSED METHOD USING DIFFERENT VALUES OF α IN THE LOSS FUNCTION (β AND γ ARE FIXED AS 0.0006 AND 0.00025, RESPECTIVELY)

Metrics	0.0005	0.005	0.05	0.5	5
Q_{MI} [46] ↑	0.4148	0.4063	**0.4401**	0.4338	0.3939
Q_{NCIE} [47] ↑	0.8075	0.8074	**0.8078**	0.8037	0.8067
Q_P [48] ↑	0.3623	0.3542	**0.3938**	0.3641	0.3096
$MS\text{-}SSIM$ [49] ↑	0.6822	0.6652	**0.6884**	0.6634	0.6574
Q_{CV} [50] ↓	127.0317	72.6817	**69.2637**	167.9059	112.4705

β

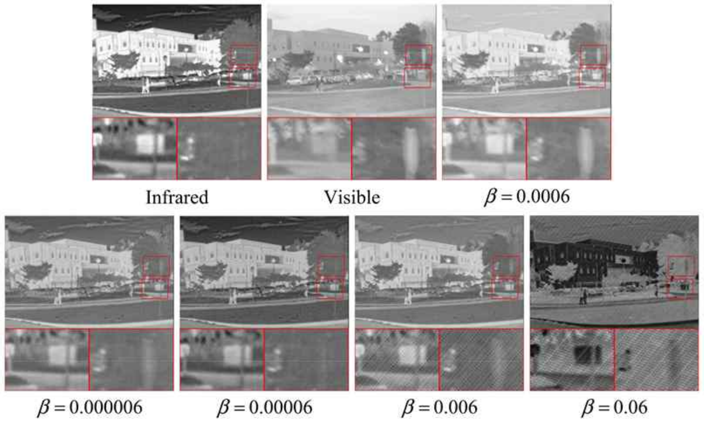
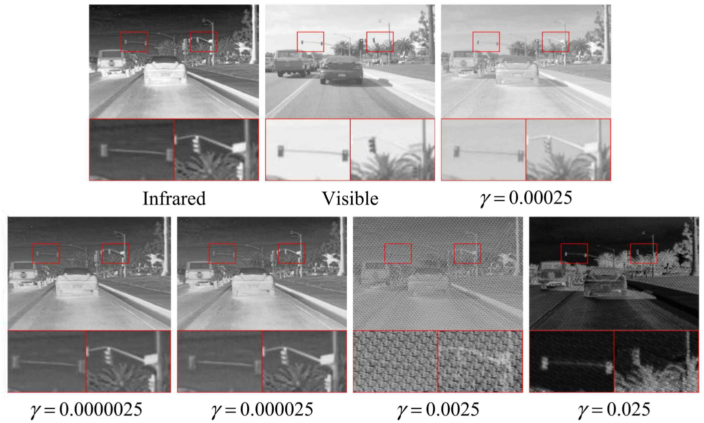


Fig. 5. A pair of source images and their corresponding fusion results of the proposed method using different values of β in the loss function (α and γ are fixed as 0.05 and 0.00025, respectively).

TABLE IV
OBJECTIVE EVALUATION OF THE PROPOSED METHOD USING DIFFERENT VALUES OF β IN THE LOSS FUNCTION (α AND γ ARE FIXED AS 0.05 AND 0.00025, RESPECTIVELY)

Metrics	0.00006	0.00006	0.0006	0.006	0.06
Q_{MI} [46] ↑	0.4267	0.4329	**0.4401**	0.194	0.1764
Q_{NCIE} [47] ↑	0.8069	0.8024	**0.8078**	0.8039	0.8027
Q_P [48] ↑	0.344	0.3199	**0.3938**	0.1541	0.0764
$MS\text{-}SSIM$ [49] ↑	0.6733	0.6512	**0.6884**	0.6005	0.5996
Q_{CV} [50] ↓	164.2465	179.7848	**69.2637**	115.9945	364.4186

γ



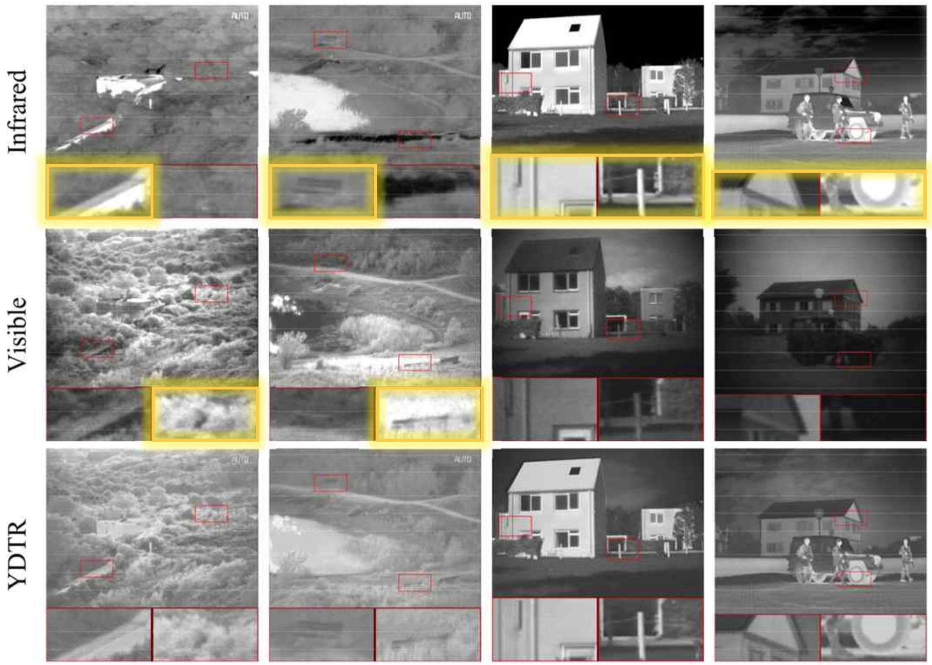
g. 6. A pair of source images and their corresponding fusion results of the oposed method using different values of γ in the loss function (α and β are ‹ed as 0.05 and 0.0006, respectively).

TABLE V
OBJECTIVE EVALUATION OF THE PROPOSED METHOD USING DIFFERENT VALUES OF γ IN THE LOSS FUNCTION (α AND β ARE FIXED AS 0.05 AND 0.0006, RESPECTIVELY)

Metrics	0.000025	0.000025	0.00025	0.0025	0.025
Q_{MI} [46] ↑	0.4344	0.4248	**0.4401**	0.11	0.2443
Q_{NCIE} [47] ↑	0.8043	0.8041	**0.8078**	0.8032	0.8034
Q_P [48] ↑	0.374	0.3385	**0.3938**	0.0504	0.1068
$MS\text{-}SSIM$ [49] ↑	0.6734	0.6671	**0.6884**	0.39	0.3206
Q_{CV} [50] ↓	166.6168	198.5401	**69.2637**	332.6176	462.2035

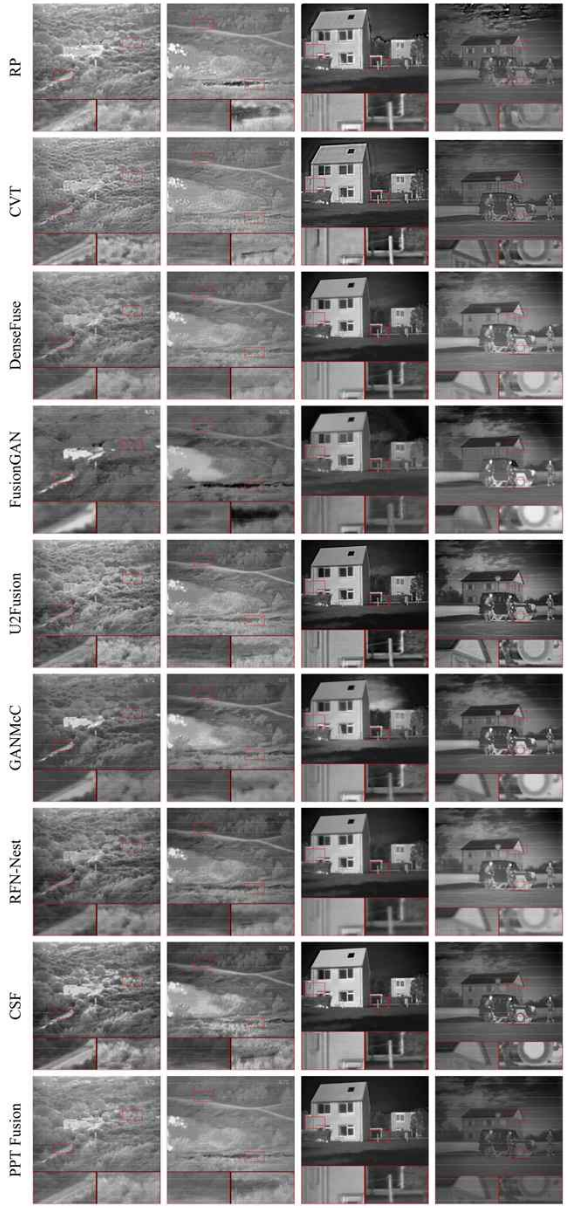
Experiment and results

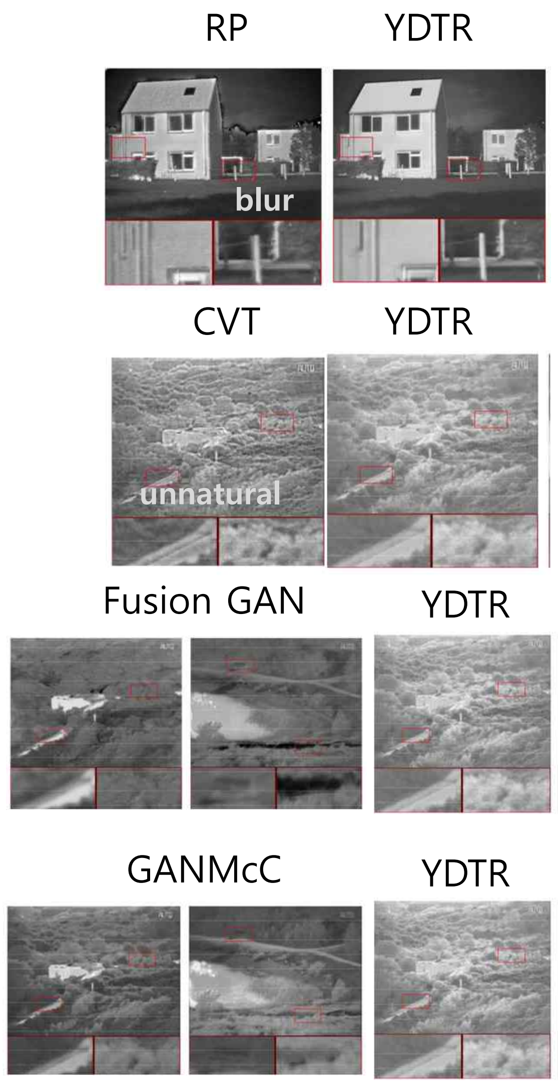
Dataset



	Q_{MI} [46] ↑	Q_{NCIE} [47] ↑	Q_P [48] ↑	$MS\text{-}SSIM$ [49] ↑	Q_{CV} [50] ↓
RP [8]	0.2449	0.8038	0.2498	0.6400	61.1259
CVT [9]	0.2420	0.8037	0.2721	0.6572	<u>45.7575</u>
DenseFuse [18]	<u>0.3390</u>	0.8050	<u>0.2936</u>	0.6769	47.9642
FusionGAN [19]	0.3335	<u>0.8050</u>	0.1025	0.6123	98.5990
U2Fusion [20]	0.2936	0.8045	0.2815	0.6594	49.6913
GANMcC [24]	0.3314	0.8047	0.2359	0.6619	65.3504
RFN-Nest [23]	0.2979	0.8045	0.2506	0.6687	54.5542
CSF [29]	0.2956	0.8045	0.2561	0.6765	48.0904
PPT Fusion [32]	0.3162	0.8047	0.2691	**0.6902**	64.0136
YDTR	**0.3526**	**0.8053**	**0.2978**	<u>0.6827</u>	**41.5507**

The mean values of 20 image pairs from the TNO dataset are listed. For each metric, the optional and suboptional values are labeled in bold and underlined, Respectively.



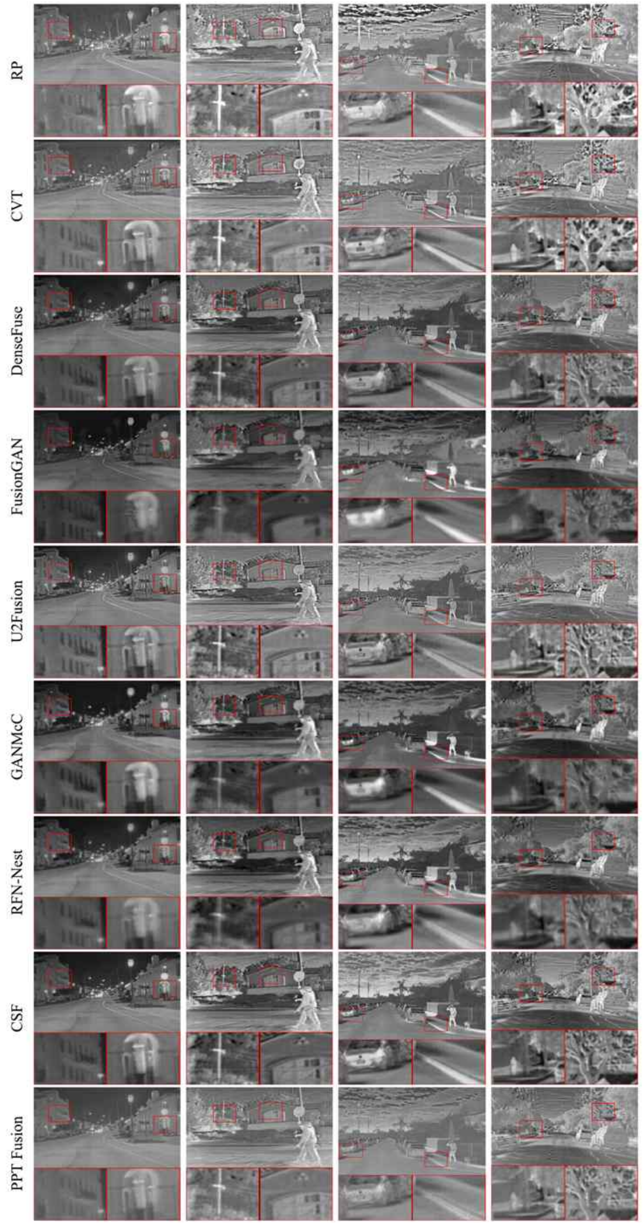


Experiment and results

Dataset

	Q_{MI} [46] ↑	Q_{NCIE} [47] ↑	Q_P [48] ↑	$MS\text{-}SSIM$ [49] ↑	Q_{CV} [50] ↓
RP [8]	0.3035	0.8054	0.3322	0.6128	139.9316
CVT [9]	0.2873	0.8051	0.3119	0.6282	116.8517
DenseFuse [18]	0.3943	0.8070	0.3520	**0.6712**	95.0244
FusionGAN [19]	0.3800	0.8068	0.1220	0.5692	138.8463
U2Fusion [20]	0.3403	0.8059	0.3349	0.6597	96.0016
GANMcC [24]	0.3620	0.8062	0.2791	0.6373	111.9141
RFN-Nest [23]	0.3618	0.8063	0.2521	0.6585	119.4961
CSF [29]	0.3668	0.8065	0.3356	0.6596	<u>92.3366</u>
PPT Fusion [32]	<u>0.4030</u>	**0.8072**	<u>0.3546</u>	<u>0.6682</u>	128.9598
YDTR	**0.4035**	<u>0.8070</u>	**0.3592**	0.6653	**60.8876**

The mean values of 20 image pairs from the roadscene data set are listed. For each metric, the optional and suboptional values are labeled in bold and underlined, respectively.



CSF YDTR

PPT Fusion

Experiment and results

Efficiency Study

RUNNING TIME OF DIFFERENT METHODS (UNIT: SECONDS)

Method	TNO Dataset	RoadScene Dataset
RP [8]	**0.0742**	**0.0483**
CVT [9]	0.7086	0.4167
DenseFuse [18]	0.5663	0.3190
FusionGAN [19]	2.6796	1.1442
U2Fusion [20]	2.2085	1.1639
GANMcC [24]	5.6752	2.3813
RFN-Nest [23]	2.3096	0.9423
CSF [29]	10.3311	5.5395
PPT Fusion [32]	0.4126	0.2203
YDTR	<u>0.1284</u>	<u>0.1626</u>

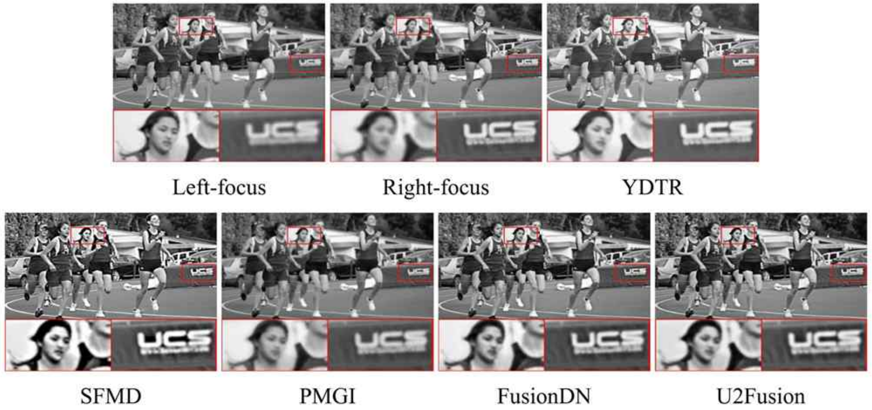
EFFICIENCY COMPARISON ON PARAMETERS AND FLOPS

Method	Paramters (M)	FLOPs (G)
DenseFuse [18]	<u>0.8903</u>	6.7291
FusionGAN [19]	5.3056	0.5469
U2Fusion [20]	2.6369	7.8948
GANMcC [24]	9.1022	1.0237
RFN-Nest [23]	19.1660	7.6763
CSF [29]	1.6397	**0.0004**
PPT Fusion [32]	312.5742	52.0814
YDTR	**0.8711**	<u>0.2113</u>

Experiment and results

Extension Study

Multi-focus image fusion w/o fine-tuning



Metrics	SFMD	PMGI	FusionDN	U2Fusion	YDTR
Q_{MI} [46] ↑	0.6732	<u>0.8614</u>	0.7259	0.7166	**0.9119**
Q_{NCIE} [47] ↑	0.8204	<u>0.8272</u>	0.8221	0.8215	**0.8287**
Q_P [48] ↑	0.7694	0.771	0.7722	**0.8121**	<u>0.7947</u>
Q_{CV} [50] ↓	25.3231	<u>5.5139</u>	10.1434	7.7625	**4.2887**
$MS\text{-}SSIM$ [49] ↑	0.9319	0.96	0.9513	<u>0.963</u>	**0.9727**

The mean values of 20 testing samples are listed. For each metric, the optional and suboptimal values are labeled in bold and underlined, respectively.

Infrared-RGB Visible image fusion

	Q_{MI} [46] ↑	Q_{NCIE} [47] ↑	Q_P [48] ↑	$MS\text{-}SSIM$ [49] ↑	Q_{CV} [50] ↓
RP [8]	0.2784	0.8036	0.2711	0.6989	124.4088
CVT [9]	0.2941	0.8042	0.3418	0.7293	<u>54.73474</u>
DenseFuse [18]	<u>0.4322</u>	<u>0.8066</u>	**0.3981**	**0.7588**	70.5651
FusionGAN [19]	0.3390	0.804	0.1359	0.7018	258.8455
U2Fusion [20]	0.3855	0.8056	0.3316	0.7456	70.4063
GANMcC [24]	0.4164	0.8059	0.3218	0.7328	97.5859
RFN-Nest [23]	0.3936	0.8058	0.3434	0.7189	80.9327
CSF [29]	0.3973	0.8054	0.3391	0.7553	82.7246
PPT Fusion [32]	0.3503	0.8043	0.3091	<u>0.758</u>	149.4944
YDTR	**0.4566**	**0.8076**	<u>0.3800</u>	0.7293	**51.8019**

The mean values of 361 image pairs from the MSRS dataset are listed. For each metric, the optional and suboptimal values are labeled in bold and underlined, respectively.

Conclusion

- End-to-End infrared and visible image fusion method via a Y-shape dynamic Transformer
- Y-branch network extracts thermal radiation information and texture details separately.
- DTRM is designed to ensure the fully excavate local useful information and global features.
- A loss function comprised of an SSIM term and an SF term is devised.

- Effectiveness of the proposed network are revealed through the objective metrics.
- Further, the proposed model can adjust to the Infrared and RGB-visible image fusion and multi-focus image fusion tasks without fine-tuning.

- In overall, using objective indicators is good to see the effectiveness, but there is no clear differences in the results shown in the figures.
- The importance of Infrared features and Visible features changes for each task, but there is no mention of this.

Question

Thank you